TEAM HAC

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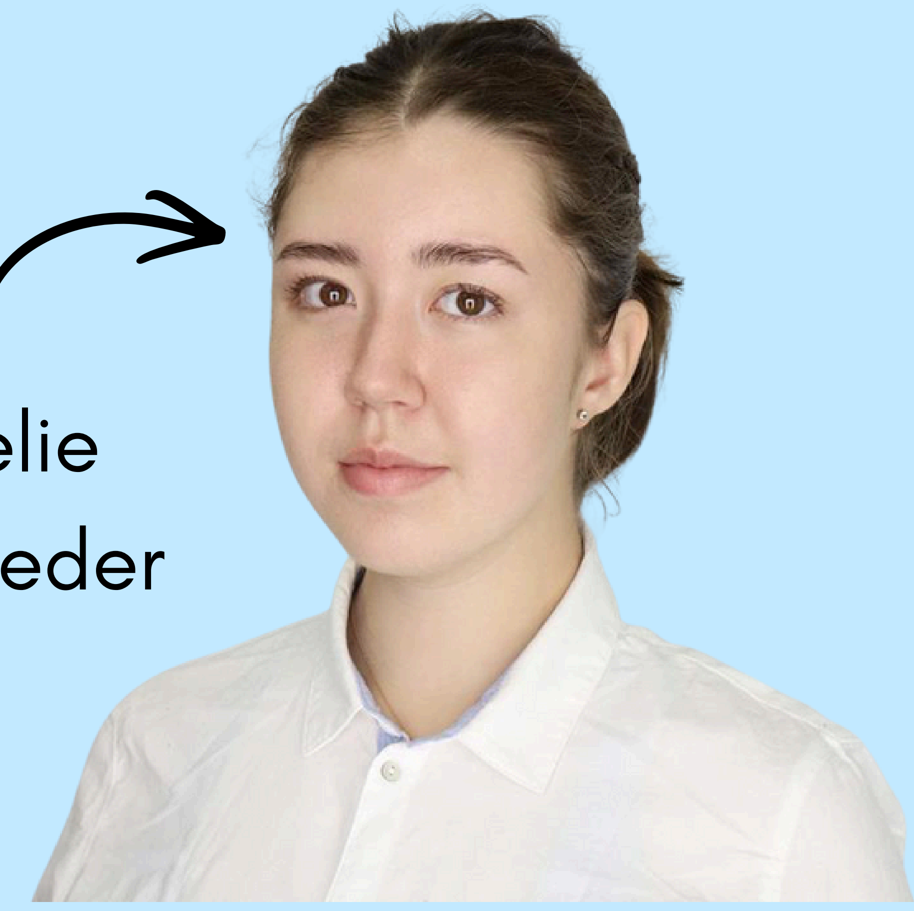


Amelie Schroeder

ISSUES

- How can we estimate patient-specific benefits and risks of surgery vs conservative care to support shared decision-making?
- Why it matters:
  - Average treatment effects hide clinical heterogeneity
  - The same surgery helps some patients but harms others

DATA

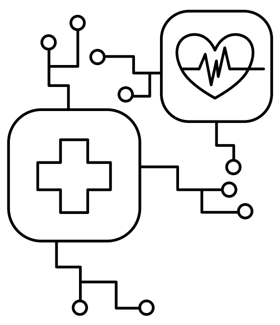
- Surgery rate ≈ 24%, target imbalance
- Outcomes skewed & heterogeneous
- Missing data
- Missing data description

ANALYTICS

- Propensity score weighting
- Uplift-aligned F-filter
- Causal meta-learners (S, T, X) with MLP and XG Boost
- Doubly-robust learning (R-learner)
- 5-fold Cross-Validation R-loss
- Feature Attribution on Treatment Effects

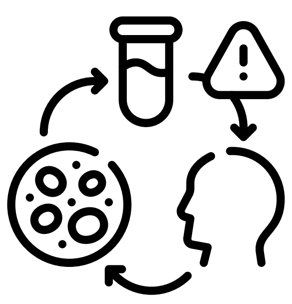
MODEL

- Across all 5 outcomes, S-learner with XGBoost model achieved the lowest cross-fitted R-loss, with stable factual MAE/RMSE and improved covariate balance (73% features balanced after IPW).



Y1: S-learner | R-loss = 0.003784 ± 0.000033
Y2: S-learner | R-loss = 0.003626 ± 0.000040
Y3: S-learner | R-loss = 0.003767 ± 0.000029
Y4 (Cost): S-learner | R-loss = 2.2 ± 0.09
Y5 (Rehab weeks): S-learner | R-loss = 2.238164 ± 0.006805

- **Impact:** Reveals patient-level treatment differences, enabling personalized decisions beyond population averages.
- **Improvement:** Can be extended with richer data, external validation or for different disease, and EHR integration to improve real-world use.



TAKEHOME/ RELEVANCE

- Provides patient-specific estimates of benefits, risks, costs, and recovery time to support shared decision-making.
- This approach shifts clinical decision-making from "one-size-fits-all" to data-driven, patient-centered care.